[redacted]

[redacted] | [redacted]@gmail.com | linkedin.com/in/[redacted] | github.com/[redacted] | [redacted]

Education

[redacted] [redacted]
Bachelor of Technology in Information Technology *Expected Graduation: May 2027*
Relevant Coursework: Data Structures & Algorithms, Data Science, Cloud Computing, Software Engineering

Experience

Software Engineer Intern Sept 2025 – Feb 2026
[redacted] [redacted]
- Re-engineered the company website in Next.js with a modular architecture, enabling support for 5+ languages
- Designed a deterministic recommendation engine using weighted linear scoring, replacing legacy heuristics
- Built a DynamoDB-backed retention pipeline triggering personalized re-engagement via SES
- Resolved Google/LINE OAuth authentication bugs to enable successful beta launch with working social login
- Co-developed a churn prediction classifier using simulated user behaviour data to establish baseline risk metrics

Undergraduate Research Assistant Jan 2026 – Present
[redacted] [redacted]
- Designed hybrid RL web crawler combining Q-Learning, Contextual Bandits, and GNNs; bootstrapped 2,153-node graph with 12,399 edges across 3 topic domains
- Built feature extraction pipeline (174 dims) for RL crawler; bootstrapped dataset with 17.4% baseline harvest rate, targeting 3-4x improvement via adaptive learning

Student Researcher July 2025 – Nov 2025
[redacted] [redacted]
- Expanded training data by 50x (850 to 42.6k) samples using synthetic augmentation to counter data scarcity
- Fine-tuned ResNet50 classifier achieving 90% accuracy on dry fruit quality grading across 3 categories, employing early stopping to reduce training time
- Deployed model to Hugging Face Spaces using Streamlit, creating an interactive UI for real-time quality grading

Projects

VeriFact (Hackathon – Ranked 7th/30) | *Python, TensorFlow, PostgreSQL*
- Built an automated fact-checking system using RAG and NLI to verify claims with evidence-backed verdicts
- Designed caching layer to store Google Search results in PostgreSQL, reducing repeat query latency and minimizing redundant API calls for the fact-checking pipeline

MyRedis | *Go, Bash*
- Engineered an in-memory key-value store supporting RESP protocol and core Redis commands (HSET, GET, DEL)
- Architected a concurrent request handler using Goroutines and Mutexes to enable multi-core scaling
- Implemented AOF persistence via buffered file I/O, ensuring data durability and crash recovery with minimal latency

WatcherX | *TypeScript, Node.js, Express.js, GCP, Firebase, Docker*
- Developed a REST API with Cloud Functions to handle authenticated video uploads
- Deployed a containerized Cloud Run service to automatically downscale videos, reducing file size by 70%
- Used Firestore real-time status tracking to provide live progress updates during video processing

Agentic Firewall | *Python*
- Built a security middleware proxy to validate AI agent tool calls, preventing unauthorized file system access
- Enforced strict filesystem controls via path normalization to mitigate directory traversal risks
- Implemented a JSON-driven policy engine to execute granular allow-lists and threat logging

Technical Skills

Languages: Python, C++, TypeScript, Go
Frameworks/Tools: Node.js, Express.js, Next.js, TensorFlow, Docker, Git, Bash, Ubuntu
Cloud & Databases: AWS, GCP, Firebase, PostgreSQL, DynamoDB